

Introduction to Deep Learning

Session 8: Classification & Maximum Likelihood

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Lecture Overview

1. Classification Problem

2. Binary Classification

3. Multiclass Classification

4. Maximum Likelihood

From Regression to Classification

Regression: predict continuous value $y \in \mathbb{R}$

- Output: house price, temperature, critical temperature
- Loss: MSE — penalises distance between prediction and target

Classification: predict discrete category $y \in \{1, \dots, K\}$

- Output: image class, spam or not, medical diagnosis
- Output space fundamentally different — no notion of distance between classes

Problem: raw network output $f_{\theta}(\mathbf{x}) \in \mathbb{R}$ cannot be interpreted as class directly

Idea: convert output to probability distribution over classes

Requirements for valid probability distribution over K classes:

$$p_k \geq 0 \quad \forall k \quad \text{and} \quad \sum_{k=1}^K p_k = 1$$

Raw network outputs (logits) $\mathbf{z} = f_{\theta}(\mathbf{x}) \in \mathbb{R}^K$

Solution: transformation that maps logits to valid probability distribution

- **Binary classification** ($K = 2$): **sigmoid** → next section
- **Multiclass classification** ($K > 2$): **softmax** → following section

Loss function: for probability outputs? not MSE, hm ...

Binary Classification

Binary classification: $y \in \{0, 1\}$ - network outputs single logit $z = f_{\theta}(\mathbf{x}) \in \mathbb{R}$

Sigmoid: squashes logit to $(0, 1)$ - interpreted as probability of class 1

$$\hat{p} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

- $\sigma(z) \in (0, 1)$ for all $z \in \mathbb{R}$
- $\sigma(0) = 0.5$ - symmetric around origin
- $\sigma(z) \rightarrow 1$ as $z \rightarrow \infty$
- $\sigma(z) \rightarrow 0$ as $z \rightarrow -\infty$

sigmoid

Loss: penalise confident wrong predictions

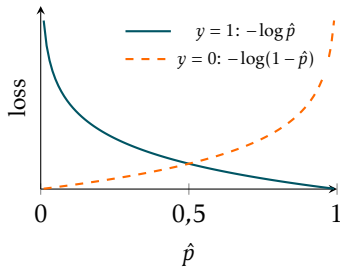
$$\ell(\hat{p}, y) = -y \log \hat{p} - (1 - y) \log(1 - \hat{p})$$

$y = 1$: loss = $-\log \hat{p}$

- $\hat{p} \rightarrow 1$: loss $\rightarrow 0$ ✓
- $\hat{p} \rightarrow 0$: loss $\rightarrow \infty$ ✗

$y = 0$: loss = $-\log(1 - \hat{p})$

- $\hat{p} \rightarrow 0$: loss $\rightarrow 0$ ✓
- $\hat{p} \rightarrow 1$: loss $\rightarrow \infty$ ✗



Training loss:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \left[y^{(i)} \log \hat{p}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{p}^{(i)}) \right]$$

Multiclass Classification

Multiclass classification: $y \in \{1, \dots, K\}$ - network outputs K logits $\mathbf{z} \in \mathbb{R}^K$

Softmax: maps logits to valid probability distribution

$$\hat{p}_k = \frac{e^{z_k}}{\sum_{j=1}^K e^{z_j}}, \quad k = 1, \dots, K$$

- $\hat{p}_k > 0$ for all k — all probabilities positive
- $\sum_{k=1}^K \hat{p}_k = 1$ — valid distribution
- Larger logit $\Rightarrow$ larger probability

Connection to sigmoid: softmax with $K = 2$ reduces to sigmoid

$$\hat{p}_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{-(z_1 - z_2)}} = \sigma(z_1 - z_2)$$

Single logit $z = z_1 - z_2$ — one degree of freedom is redundant

Categorical Cross-Entropy

Labels: one-hot vector $\mathbf{y} \in \{0, 1\}^K$ — $y_k = 1$ for correct class, 0 otherwise

Loss: picks out log probability of correct class

$$\ell(\hat{\mathbf{p}}, \mathbf{y}) = - \sum_{k=1}^K y_k \log \hat{p}_k = -\log \hat{p}_y$$

- Correct class predicted confidently $\hat{p}_y \rightarrow 1$: loss $\rightarrow 0$ ✓
- Correct class predicted with low confidence $\hat{p}_y \rightarrow 0$: loss $\rightarrow \infty$ ✗

Training loss:

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log \hat{p}_{y^{(i)}}^{(i)}$$

Maximum Likelihood

Maximum Likelihood Principle

Setup: model defines conditional distribution $p(y \mid \mathbf{x}; \theta)$ over outputs

Likelihood: probability of observing training data under the model

$$\mathcal{L}(\theta) = \prod_{i=1}^N p(y^{(i)} \mid \mathbf{x}^{(i)}; \theta)$$

MLE: find parameters that make observed data most probable

$$\theta^* = \arg \max_{\theta} \mathcal{L}(\theta)$$

Log-likelihood: sum instead of product — same maximiser, more convenient

$$\log \mathcal{L}(\theta) = \sum_{i=1}^N \log p(y^{(i)} \mid \mathbf{x}^{(i)}; \theta)$$

Maximise log-likelihood $\equiv$ **minimise** negative log-likelihood (NLL)

$$\theta^* = \arg \max_{\theta} \sum_i \log p(y^{(i)} | \mathbf{x}^{(i)}; \theta) = \arg \min_{\theta} \underbrace{-\frac{1}{N} \sum_i \log p(y^{(i)} | \mathbf{x}^{(i)}; \theta)}_{\mathcal{L}(\theta)}$$

Key insight: choice of $p(y | \mathbf{x}; \theta)$ determines the loss function

Distributional assumption	Loss function
Gaussian	MSE
Bernoulli	Binary cross-entropy
Categorical	Categorical cross-entropy

Gaussian Assumption $\rightarrow$ MSE

Assume: $p(y | \mathbf{x}; \theta) = \mathcal{N}(y; f_{\theta}(\mathbf{x}), \sigma^2)$

$$\log p(y | \mathbf{x}; \theta) = -\frac{1}{2\sigma^2}(y - f_{\theta}(\mathbf{x}))^2 - \frac{1}{2}\log(2\pi\sigma^2)$$

NLL:

$$-\frac{1}{N} \sum_{i=1}^N \log p(y^{(i)} | \mathbf{x}^{(i)}; \theta) = \frac{1}{2\sigma^2} \underbrace{\frac{1}{N} \sum_{i=1}^N (y^{(i)} - f_{\theta}(\mathbf{x}^{(i)}))^2}_{\text{MSE}} + \text{const}$$

Minimising NLL under Gaussian noise assumption $\equiv$ minimising **MSE**

Connects to Session 3: MSE is not arbitrary — it is MLE under Gaussian noise

Bernoulli Assumption $\rightarrow$ Binary Cross-Entropy

Assume: $p(y | \mathbf{x}; \theta) = \text{Bernoulli}(y; \hat{p}), \quad \hat{p} = \sigma(f_{\theta}(\mathbf{x}))$

$$p(y | \mathbf{x}; \theta) = \hat{p}^y (1 - \hat{p})^{1-y}$$

Log-likelihood:

$$\log p(y | \mathbf{x}; \theta) = y \log \hat{p} + (1 - y) \log(1 - \hat{p})$$

NLL:

$$-\frac{1}{N} \sum_{i=1}^N \log p(y^{(i)} | \mathbf{x}^{(i)}; \theta) = -\frac{1}{N} \sum_{i=1}^N \underbrace{\left[y^{(i)} \log \hat{p}^{(i)} + (1 - y^{(i)}) \log(1 - \hat{p}^{(i)}) \right]}_{\text{binary cross-entropy}}$$

Minimising NLL under Bernoulli assumption $\equiv$ minimising **binary cross-entropy**

Categorical Assumption $\rightarrow$ Categorical Cross-Entropy

Assume: $p(y | \mathbf{x}; \theta) = \text{Categorical}(y; \hat{\mathbf{p}})$, $\hat{\mathbf{p}} = \text{softmax}(f_{\theta}(\mathbf{x}))$

$$p(y | \mathbf{x}; \theta) = \prod_{k=1}^K \hat{p}_k^{y_k} = \hat{p}_y$$

Log-likelihood:

$$\log p(y | \mathbf{x}; \theta) = \sum_{k=1}^K y_k \log \hat{p}_k = \log \hat{p}_y$$

NLL:

$$-\frac{1}{N} \sum_{i=1}^N \log p(y^{(i)} | \mathbf{x}^{(i)}; \theta) = \underbrace{-\frac{1}{N} \sum_{i=1}^N \log \hat{p}_{y^{(i)}}^{(i)}}_{\text{categorical cross-entropy}}$$